

Problem 1 (Linear Algebra Review, 30 points)

(a) (8 points)

$$\begin{aligned}
(\mathbf{A}\mathbf{v})^T &= \left(\begin{bmatrix} \mathbf{a}_1^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix} \mathbf{v} \right)^T \\
&= \begin{bmatrix} \mathbf{a}_1^T \mathbf{v} \\ \vdots \\ \mathbf{a}_n^T \mathbf{v} \end{bmatrix}^T \\
&= [\mathbf{a}_1^T \mathbf{v} \ \dots \ \mathbf{a}_n^T \mathbf{v}] \\
&= [\mathbf{v}^T \mathbf{a}_1 \ \dots \ \mathbf{v}^T \mathbf{a}_n] \\
&= \mathbf{v}^T [\mathbf{a}_1 \ \dots \ \mathbf{a}_n] \\
&= \mathbf{v}^T \mathbf{A}^T
\end{aligned}$$

(b) (7 points) Using the chain rule,

$$\begin{aligned}
\frac{\partial f(x, y)}{\partial w} &= \frac{\partial f(x, y)}{\partial x} \frac{\partial x}{\partial w} + \frac{\partial f(x, y)}{\partial y} \frac{\partial y}{\partial w} \\
&= (2x + 2y)(2w - 3) + (2x - 2y)(4w^3 - 6w + 1) \\
&= -8w^7 + 48w^5 - 40w^4 - 56w^3 + 60w^2 + 4w.
\end{aligned}$$

(c) (7 points) By the definition of the ℓ_2 norm,

$$\|\mathbf{U}\mathbf{x}\|_2^2 = (\mathbf{U}\mathbf{x})^T (\mathbf{U}\mathbf{x}) = \mathbf{x}^T \mathbf{U}^T \mathbf{U} \mathbf{x} = \mathbf{x}^T \mathbf{I} \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|_2^2$$

(d) (8 points) We can first write our $f(\mathbf{x})$ as an expression of scalars,

$$\begin{aligned}
f(\mathbf{x}) &= \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{12} & a_{22} & \dots & a_{2n} \\ & & \ddots & \\ a_{1n} & a_{2n} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \\
&= \begin{bmatrix} \sum_i a_{1i} x_i & \sum_i a_{2i} x_i & \dots & \sum_i a_{ni} x_i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \\
&= \sum_{1 \leq i \leq n} \sum_{1 \leq j \leq n} a_{ij} x_i x_j,
\end{aligned}$$

To calculate $\frac{\partial f(\mathbf{x})}{\partial x_1}$, we should only focus on the term containing x_1 in $f(\mathbf{x})$. We could rewrite $f(\mathbf{x})$ as :

$$f(\mathbf{x}) = a_{11}x_1^2 + 2 \left(\sum_{2 \leq i \leq n} a_{1i} x_i \right) x_1 + C(x_2, \dots, x_n),$$

where $C(x_2, \dots, x_n)$ is a function irrelevant of x_1 , so:

$$\frac{\partial f(\mathbf{x})}{\partial x_1} = 2 \sum_{1 \leq i \leq n} a_{1i} x_i,$$

The gradient of $f(\mathbf{x})$ w.r.t. $\mathbf{x}$,

$$\begin{aligned} \nabla_{\mathbf{x}} f(\mathbf{x}) &= \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix} = 2 \begin{bmatrix} \sum_{1 \leq i \leq n} a_{1i} x_i \\ \sum_{1 \leq i \leq n} a_{2i} x_i \\ \vdots \\ \sum_{1 \leq i \leq n} a_{ni} x_i \end{bmatrix} \\ &= 2\mathbf{A}\mathbf{x} \end{aligned}$$

Problem 2 (Probability Review, 20 points)

- (a) (10 points) Since all possible draws of 6 balls from the box have equal probability, the probability of getting exactly k matches is

$$P(k \text{ matches}) = \frac{\# \text{ of possible draws with exactly } k \text{ matches}}{\# \text{ of possible draws}}.$$

There are 6 balls to be chosen from the box. There are 51 unique balls in the box among which, for any player, there are exactly 6 balls matching the numbers on their ticket.

There are $\binom{51}{6}$ total ways to draw 6 balls from the box of 51 balls, so the denominator above is $\binom{51}{6}$. To have $k \in \{0, 1, \dots, 6\}$ matches, k of the 6 matching numbers must be drawn from the box, while the remaining $6 - k$ balls drawn from the box must come from the set of $51 - 6 = 45$ numbers that do not match the player's ticket. The number of ways that k of the 6 matching numbers can be selected is $\binom{6}{k}$, and the number of ways $6 - k$ of the 45 non-matching balls can be selected is $\binom{45}{6-k}$. Notice that any of the $\binom{6}{k}$ ways to pick k matching numbers can happen with any of the $\binom{45}{6-k}$ ways of choosing $6 - k$ non-matching balls. Then the total number of draws with exactly k matches is $\binom{6}{k} \binom{45}{6-k}$. Combining with what we said earlier, the probability of getting exactly k matches is

$$P(k \text{ matches}) = \frac{\binom{6}{k} \binom{45}{6-k}}{\binom{51}{6}}.$$

- (b) (10 points) By the law of total probability,

$$\begin{aligned} P(\text{machine lies}) &= \sum_{k=0}^6 P(\text{machine lies} \mid k \text{ matches}) P(k \text{ matches}) \\ &= \sum_{k=0}^6 P(\text{machine lies} \mid k \text{ matches}) \frac{\binom{6}{k} \binom{45}{6-k}}{\binom{51}{6}}, \end{aligned}$$

where the last step is from part (a).

We know that $P(\text{machine lies} \mid k = 0 \text{ matches}) = 0$. For $1 \leq k \leq 6$, the machine rolls k (6-sided, fair) dice, and lies if the sum of the roll is equal to k , which can only happen if all ones are rolled. The total number of possible dice rolls with k dice is 6^k , all of which are equally probable, and only one of these possible rolls will result in the machine lying. Therefore, $P(\text{machine lies} \mid k \text{ matches}) = \frac{1}{6^k}$. Putting it all together yields

$$P(\text{machine lies}) = \sum_{k=1}^6 \frac{\binom{6}{k} \binom{45}{6-k}}{6^k \binom{51}{6}}.$$

Problem 3 (Linear Least Squares, 30 points + 10 bonus points)

(a) (15 points)

$$\begin{aligned} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 &= \|\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T\mathbf{x} - \mathbf{y}\|_2^2 \\ &= \|\mathbf{Q}^T(\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T\mathbf{x} - \mathbf{y})\|_2^2 \\ &= \|\mathbf{\Lambda}\mathbf{Q}^T\mathbf{x} - \mathbf{Q}^T\mathbf{y}\|_2^2 \\ &= \|\mathbf{\Lambda}\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\|_2^2, \end{aligned}$$

where $\tilde{\mathbf{x}} \triangleq \mathbf{Q}^T\mathbf{x}$ and $\tilde{\mathbf{y}} \triangleq \mathbf{Q}^T\mathbf{y}$. The first step uses the eigendecomposition, the second step uses unitary invariance of the Euclidean norm, and the third step uses properties of unitary matrices.

(b) (15 points) Using the result from part (a) and the fact that $\mathbf{\Lambda}$ is a diagonal matrix,

$$\begin{aligned} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 &= \|\mathbf{\Lambda}\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\|_2^2 \\ &= \left\| \begin{bmatrix} \lambda_1 \tilde{x}_1 \\ \vdots \\ \lambda_n \tilde{x}_n \end{bmatrix} - \begin{bmatrix} \tilde{y}_1 \\ \vdots \\ \tilde{y}_n \end{bmatrix} \right\|_2^2 \\ &= \sum_{i: \lambda_i \neq 0} (\lambda_i \tilde{x}_i - \tilde{y}_i)^2 + \sum_{i: \lambda_i = 0} (\tilde{y}_i)^2. \end{aligned}$$

Each term of the first sum above is nonnegative, so this quantity is minimized (and is exactly zero) when we take $\tilde{x}_i = \frac{1}{\lambda_i} \tilde{y}_i$. The terms in the second sum do not depend on $\tilde{x}_i$, so we can set them to be whatever we want without affecting the minimal value of $\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2$. Therefore, there are infinitely many solutions to this problem (as long as the matrix $\mathbf{A}$ has at least one zero eigenvalue, i.e., is rank-deficient). The so-called *minimum-norm solution* is achieved by setting the values of $\tilde{x}_i$ to zero whenever $\lambda_i = 0$. Taking the minimum-norm solution, we have $\tilde{\mathbf{x}}^* = \mathbf{\Lambda}^\dagger \tilde{\mathbf{y}}$. Recall that $\tilde{\mathbf{x}}$ is a linear function of $\mathbf{x}$ and $\tilde{\mathbf{y}}$ is a linear function of $\mathbf{y}$. This means that an optimal $\tilde{\mathbf{x}}^*$ is achievable as long as it corresponds to some optimal $\mathbf{x}^*$. In this case we can find such an $\mathbf{x}^*$, since $\tilde{\mathbf{x}} = \mathbf{Q}^T\mathbf{x}$ and $\mathbf{Q}$ has full rank (all unitary matrices have full rank). Substituting the definitions for $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$ yields

$$\mathbf{Q}^T\mathbf{x}^* = \mathbf{\Lambda}^\dagger \mathbf{Q}^T\mathbf{y}.$$

We solve for $\mathbf{x}^*$ by multiplying both sides by $\mathbf{Q}$, yielding $\mathbf{x}^* = \mathbf{Q}\mathbf{\Lambda}^\dagger \mathbf{Q}^T\mathbf{y}$.

(c) (Optional Challenge Problem, 10 bonus points)

Write the singular value decomposition of an arbitrary $\mathbf{A} \in \mathbb{R}^{n \times d}$ as $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ and $\mathbf{V} \in \mathbb{R}^{d \times d}$ are unitary and $\mathbf{\Sigma} \in \mathbb{R}^{n \times d}$ is rectangular diagonal with $\sigma_i \triangleq \Sigma_{i,i} \geq 0$ for $1 \leq i \leq \min(n, d)$. Order the singular values so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\min(n, d)} \geq 0$, and let r be the number of strictly positive ones, so $\sigma_i > 0$ for $i \leq r$ and $\sigma_i = 0$ for $r < i \leq \min(n, d)$.

Step 1: reduce to a diagonal problem. Exactly as in part (a), unitary invariance of the Euclidean norm gives

$$\begin{aligned}\|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 &= \|\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\mathbf{x} - \mathbf{y}\|_2^2 \\ &= \|\mathbf{U}^T(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^T\mathbf{x} - \mathbf{y})\|_2^2 \\ &= \|\mathbf{\Sigma}\mathbf{V}^T\mathbf{x} - \mathbf{U}^T\mathbf{y}\|_2^2 \\ &= \|\mathbf{\Sigma}\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\|_2^2,\end{aligned}$$

where $\tilde{\mathbf{x}} \triangleq \mathbf{V}^T\mathbf{x} \in \mathbb{R}^d$ and $\tilde{\mathbf{y}} \triangleq \mathbf{U}^T\mathbf{y} \in \mathbb{R}^n$. Unlike in part (a), these two vectors now live in *different* spaces, since $\mathbf{\Sigma}$ need not be square.

Step 2: split the sum. The vector $\mathbf{\Sigma}\tilde{\mathbf{x}} \in \mathbb{R}^n$ has i -th entry $\sigma_i\tilde{x}_i$ for $i \leq \min(n, d)$, and i -th entry 0 for $i > \min(n, d)$ (such rows of $\mathbf{\Sigma}$ occur only when $n > d$). Therefore

$$\begin{aligned}\|\mathbf{\Sigma}\tilde{\mathbf{x}} - \tilde{\mathbf{y}}\|_2^2 &= \sum_{i=1}^{\min(n, d)} (\sigma_i\tilde{x}_i - \tilde{y}_i)^2 + \sum_{i=\min(n, d)+1}^n \tilde{y}_i^2 \\ &= \sum_{i=1}^r (\sigma_i\tilde{x}_i - \tilde{y}_i)^2 + \sum_{i=r+1}^n \tilde{y}_i^2.\end{aligned}$$

Note that the second sum runs all the way up to n : it collects both the indices $r < i \leq \min(n, d)$ for which $\sigma_i = 0$ and, when $n > d$, the $n - d$ trailing all-zero rows of $\mathbf{\Sigma}$. Only the first sum depends on $\tilde{\mathbf{x}}$.

Step 3: minimize. Every term above is nonnegative, and the i -th term of the first sum is driven to zero by the choice $\tilde{x}_i = \tilde{y}_i/\sigma_i$. The coordinates $\tilde{x}_i$ with $r < i \leq d$ do not appear in the objective at all, so they may be set arbitrarily without changing the value of f ; setting them to zero gives the minimum-norm solution. Hence

$$\tilde{x}_i^* = \begin{cases} \frac{1}{\sigma_i}\tilde{y}_i, & 1 \leq i \leq r \\ 0, & r < i \leq d, \end{cases}$$

and the minimum value attained is $\sum_{i=r+1}^n \tilde{y}_i^2$, which is zero exactly when $\mathbf{y}$ lies in the range of $\mathbf{A}$.

Defining $\mathbf{\Sigma}^\dagger \in \mathbb{R}^{d \times n}$ by $\Sigma_{i,j}^\dagger = \begin{cases} \frac{1}{\sigma_j}, & \Sigma_{j,i} \neq 0 \\ 0, & \text{otherwise} \end{cases}$, we get $(\mathbf{\Sigma}^\dagger\tilde{\mathbf{y}})_i = \tilde{y}_i/\sigma_i$ for $i \leq r$ and 0 for

$i > r$, so the display above says precisely that $\tilde{\mathbf{x}}^* = \mathbf{\Sigma}^\dagger\tilde{\mathbf{y}}$.

Step 4: undo the change of variables. Since $\mathbf{V}$ is unitary, $\tilde{\mathbf{x}} = \mathbf{V}^T\mathbf{x}$ is a one-to-one correspondence, so an optimal $\tilde{\mathbf{x}}^*$ is attained by some optimal $\mathbf{x}^*$. Substituting the definitions gives $\mathbf{V}^T\mathbf{x}^* = \mathbf{\Sigma}^\dagger\mathbf{U}^T\mathbf{y}$, and multiplying on the left by $\mathbf{V}$ yields

$$\mathbf{x}^* = \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^T\mathbf{y}.$$

This identifies the pseudo-inverse as $\mathbf{A}^\dagger = \mathbf{V}\mathbf{\Sigma}^\dagger\mathbf{U}^T$. Specializing to a symmetric positive semidefinite $\mathbf{A}$, where one may take $\mathbf{U} = \mathbf{V} = \mathbf{Q}$ and $\mathbf{\Sigma} = \mathbf{\Lambda}$, recovers part (b).

Problem 4 (numpy Exercises, 20 points)

(a-b) See the solution notebook `q4_sol.ipynb` uploaded to Canvas.